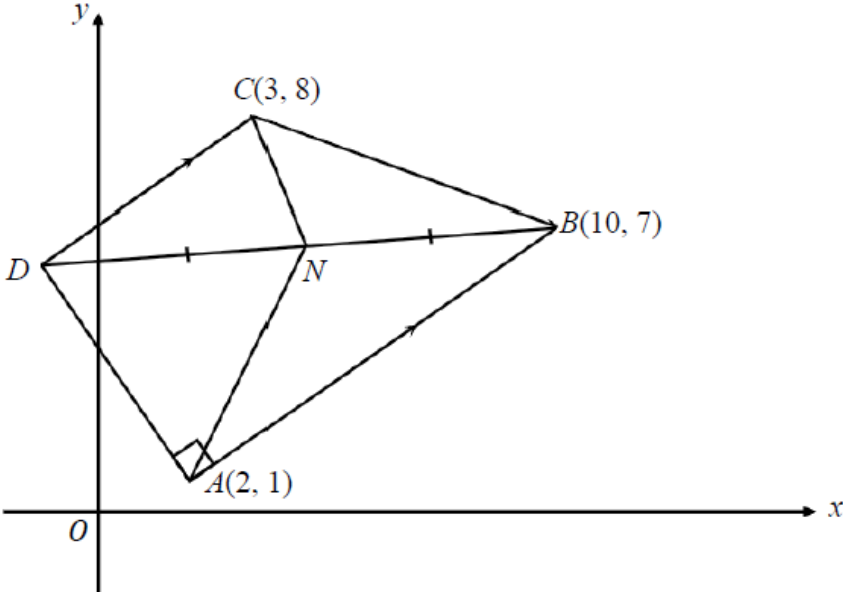
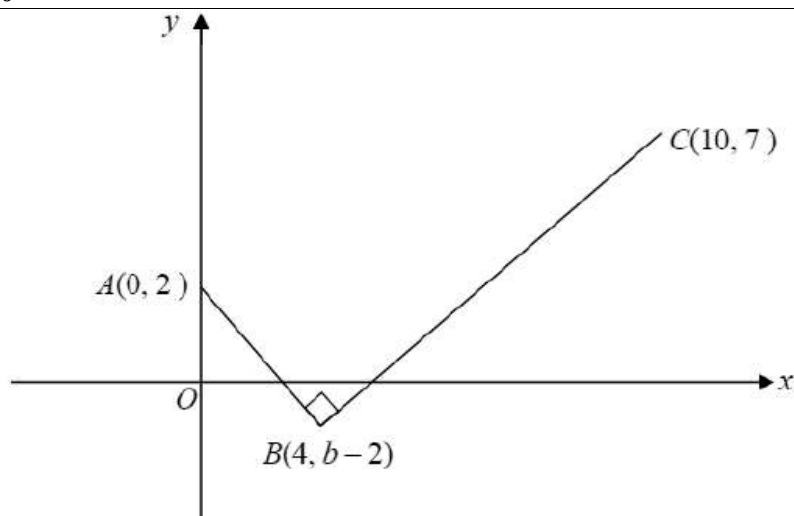


1	ACSBR/2025/I/01 Three points are given by $A(1, 4)$, $B(9, 8)$ and $C(7, 12)$. (a) Show that angle ABC is 90°. [3] (b) A circle can be drawn through A, B and C. Explain why the tangent to the circle at B is parallel to the y-axis. [3]
2	ACSBR/2025/I/13 The diagram shows a trapezium $ABCD$ with vertices $A(2, 1)$, $B(10, 7)$, $C(3, 8)$ and D. Angle BAD is 90° and N is the midpoint of BD.  (a) Show that the coordinates of D are $(-1, 5)$. [6] (b) Find the area of $ANCD$. [3]
3	ACSBR/2025/II/10 The points $A(3, -1)$ and $B(3, 9)$ lie on the circumference of a circle. The line $y = 17$ is a tangent to the circle. The x-coordinate of the centre of the circle is positive. (a) Find the radius of the circle and the coordinates of its centre. [4] (b) Hence, find the coordinates of the point on the circle which is nearest to the y-axis. [2]



The diagram shows the coordinates of three vertices of a quadrilateral $ABCD$ in which angle $ABC = 90^\circ$.

The vertices are at points $A(0, 2)$, $B(4, b-2)$ and $C(10, 7)$.

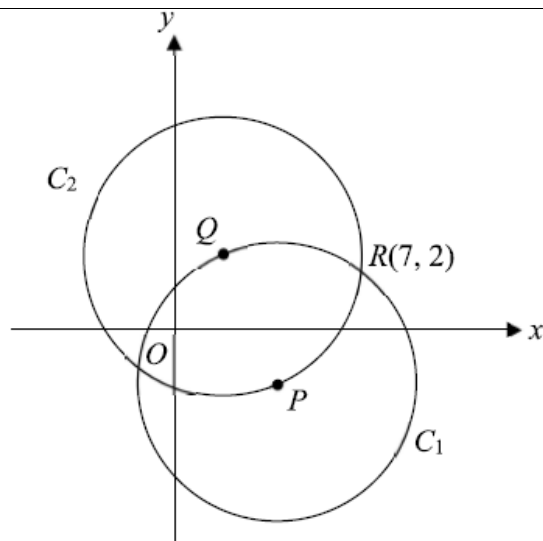
(a) Find the value of b . [3]

(b) Find the equation of the perpendicular bisector of BC . [3]

The vertex D is the point where the perpendicular bisector of BC cuts the y -axis.

(c) State the coordinates of D . [2]

(d) Calculate the area of quadrilateral $ABCD$. [2]

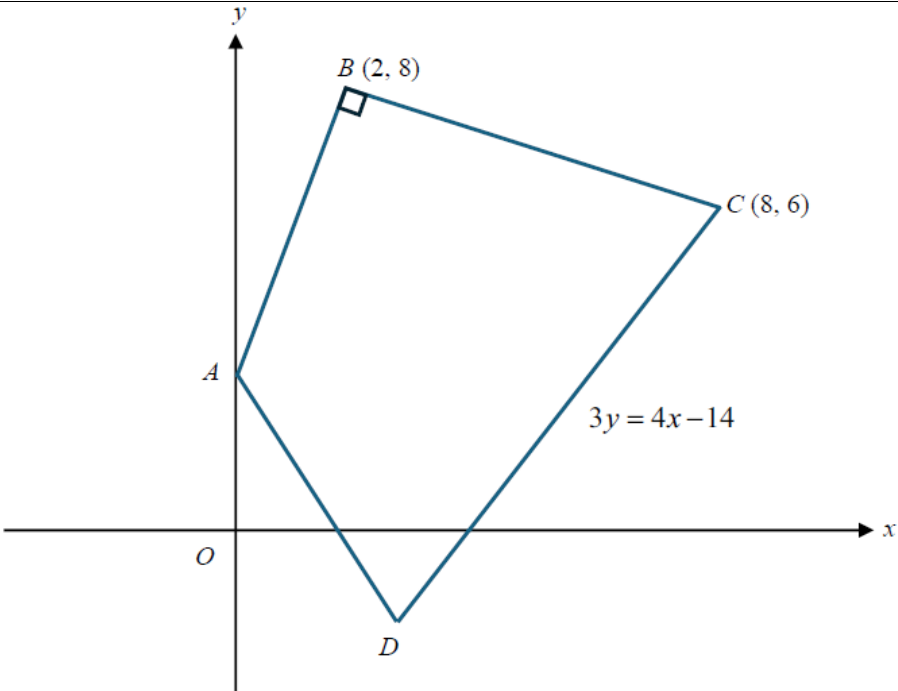


The diagram shows two intersecting circles, C_1 and C_2 , with centres P and Q respectively such that Q lies on C_1 and P lies on C_2 . The point R is $(7, 2)$ and lies on both circles.

The equation of circle C_1 is $x^2 + y^2 - 6x + 4y - 19 = 0$.

(a) Find the coordinates of P and the radius of C_1 . [4]

(b) Find the equation of the perpendicular bisector of PR . [3]

	(c) Show that the coordinates of Q are $(a + b\sqrt{3}, c + d\sqrt{3})$, where a, b, c and d are integers. [4] (d) Calculate the area of the triangle PQR. [2]
6	CCHY/2025/II/9 The equation of a circle is $x^2 + y^2 - 8x - 4y + 15 = 0$. (a) Find the radius and coordinates of the centre of the circle. [3] Two points are given by $A(5, -2)$ and $B(9, 2)$. The perpendicular bisector of AB cuts the circle at point C and D. (b) Find the coordinates of C and of D. [6] (c) Find the shortest distance from the origin O to the line segment CD, giving your answer in the form $k\sqrt{2}$, where k is a constant to be determined. [3]
7	GMS/2025/I/5  The diagram above shows a quadrilateral $ABCD$. Point B is $(2, 8)$ and point C is $(8, 6)$. The point D lies on the perpendicular bisector of BC and the point A lies on the y-axis. The equation CD is $3y = 4x - 14$ and angle $ABC = 90^\circ$. Find (a) the gradient of AB, [1] (b) the coordinates of A, [2] (c) the equation of the perpendicular bisector of BC, [2] (d) the coordinates of D. [3]

8	GMS/2025/II/4
	The equation of a circle is $x^2 + y^2 - 12x - 8y - 13 = 0$. (a) Find the radius and the coordinates of the centre, C, of the circle. [2] (b) Find the equation of the line which passes through C and is perpendicular to the line $4x + 7y = 117$. [3] (c) Show that the line $4x + 7y = 117$ is a tangent to the circle and state the coordinates where the line touches the circle. [5]

1	(a) $\text{Grad}_{AB} = \frac{8-4}{9-1} = \frac{1}{2}$ $\text{Grad}_{BC} = \frac{12-8}{7-9} = -2$ $\text{Grad}_{AB} \times \text{Grad}_{BC} = -1 \text{ } AB \text{ perpendicular to } BC.$ Hence angle $ABC = 90^\circ$ (b) Centre of circle = $\left(\frac{7+1}{2}, \frac{12+4}{2}\right) = (4, 8)$ grad of radius to the tangent = $\frac{8-8}{4-9} = 0$ grad of tangent = $-\frac{1}{0} = \text{undefined}$ Hence the tangent is parallel to the y-axis.
2	(a) $\frac{8-b}{3-a} = \frac{3}{4}$ $3a - 4b = -23$ $\frac{b-1}{a-2} = \frac{-4}{3}$ $3b + 4a = 11$ $a = -1$ $b = 5$ $D(-1, 5)$ (b) $N = \left(\frac{9}{2}, 6\right)$ $\text{Area} = \frac{1}{2} \times \begin{vmatrix} 2 & \frac{9}{2} & 3 & -1 & 2 \\ 1 & \frac{2}{6} & 8 & 5 & 1 \end{vmatrix}$ $= \frac{1}{2} \times (62 - 24.5)$ $= 18.75 \text{ sq units}$
3	(a) y coordinate of centre = $\frac{9-1}{2} = 4$ $R = 17 - 4 = 13$ $C = (a, 4)$ $(x-a)^2 + (y-4)^2 = 169$ $(3-a)^2 + (-1-4)^2 = 169$ $(3-a)^2 = 144$ $3-a = 12 \text{ or } 3-a = -12$ $a = -9 \text{ (reject, } a > 0) \text{ or } a = 15$ Centre $(15, 4)$ (b) $15 - 13 = 2$ $(2, 4)$

4	a Gradient of AB $\times$ Gradient of BC = -1 $\frac{b-2-2}{4-0} \times \frac{b-2-7}{4-10} = -1$ $\frac{b-4}{4} \times \frac{b-9}{-6} = -1$ $(b-4)(b-9) = 24$ $b^2 - 13b + 36 - 24 = 0$ $b^2 - 13b + 12 = 0$ $(b-1)(b-12) = 0$ $b = 1 \text{ or } b = 12 \text{ (rejected)}$ b Midpoint of BC = (7, 3) Gradient of perpendicular bisector = Gradient of AB = $\frac{2-(-1)}{0-4} = \frac{-3}{4}$ Equation of perpendicular bisector $y-3 = \frac{-3}{4}(x-7) \quad \text{or} \quad y = \frac{-3}{4}x + \frac{33}{4}$ c $\left(0, \frac{33}{4}\right)$ d $\frac{1}{2} \begin{vmatrix} 0 & 4 & 10 & 0 & 0 \\ 2 & -1 & 7 & \frac{33}{4} & 2 \end{vmatrix}$ $\frac{1}{2} \left 28 + \frac{330}{4} - (8-10) \right $ $= 56\frac{1}{4} \text{ units}^2$
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5	a Mtd 1 $x^2 + y^2 - 6x + 4y - 19 = 0$ $\text{Centre} \left(\frac{-6}{-2}, \frac{4}{-2} \right) = (3, -2)$ $\text{radius} = \sqrt{(3)^2 + (-2)^2 - (-19)}$ $= \sqrt{32}$ $= 4\sqrt{2}$ Mtd 2 $x^2 + y^2 - 6x + 4y - 19 = 0$ $(x-3)^2 + (y+2)^2 = 19 + (-3)^2 + (2)^2$ $(x-3)^2 + (y+2)^2 = 32$ Centre(3, -2) $\text{radius} = \sqrt{32}$ $= 4\sqrt{2}$ b $\text{Gradient of } PR = \frac{-2-2}{3-7}$ $= 1$ Gradient of perpendicular = -1 Mid-point of PR = (5, 0) Equation of perpendicular bisector is : $y - 0 = -(x - 5)$ $y = -x + 5$ c $x^2 + y^2 - 6x + 4y - 19 = 0 \quad \text{---(1)}$ $y = -x + 5 \quad \text{---(2)}$ Sub. (2) into (1), $x^2 + (-x+5)^2 - 6x + 4(-x+5) - 19 = 0$ $x^2 + x^2 - 10x + 25 - 6x - 4x + 20 - 19 = 0$ $2x^2 - 20x + 26 = 0$ $x = \frac{20 \pm \sqrt{20^2 - 4(2)(26)}}{2(2)}$ $= 5 - 2\sqrt{3} \text{ or } 5 + 2\sqrt{3} \text{ (rejected)}$ $y = 2\sqrt{3}$ $\therefore Q \text{ is } (5 - 2\sqrt{3}, 2\sqrt{3})$ d $\text{Area of } \triangle PQR = \frac{1}{2} (\sqrt{32})^2 \sin 60^\circ$ $= 8\sqrt{3} \text{ or } 13.9 \text{ units}^2$
6	(a)

$$x^2 + y^2 - 8x - 4y + 15 = 0$$

$$x^2 - 8x + 4^2 + y^2 - 4y + 2^2 = 4^2 + 2^2 - 15$$

$$(x-4)^2 + (y-2)^2 = 5$$

Radius is $\sqrt{5}$ units

Centre (4, 2)

(b)

$$m_{AB} = \frac{-2-2}{5-9}$$

$$= 1$$

$$\text{Midpoint of } AB = \left(\frac{5+9}{2}, \frac{-2+2}{2} \right)$$

$$= (7, 0)$$

Equation of perpendicular bisector is

$$y-0 = -1(x-7)$$

$$y = -x + 7$$

Substituting equation of line into equation of circle:

$$(x-4)^2 + (-x+7-2)^2 = 5$$

$$x^2 - 8x + 16 + x^2 - 10x + 25 = 5$$

$$2x^2 - 18x + 36 = 0$$

$$x^2 - 9x + 18 = 0$$

$$(x-3)(x-6) = 0$$

$$x = 3 \quad \text{or} \quad x = 6$$

$$y = 4 \quad y = 1$$

The coordinates are (3, 4) and (6, 1).

(c)

$$\text{Area of } OCD = \frac{1}{2} \begin{vmatrix} 0 & 6 & 3 & 0 \\ 0 & 1 & 4 & 0 \end{vmatrix}$$

$$= \frac{1}{2} [(0+24+0) - (0+3+0)]$$

$$= 10.5$$

$$\text{Length of } CD = \sqrt{(3-6)^2 + (4-1)^2}$$

$$= \sqrt{18}$$

$$= 3\sqrt{2}$$

$$\text{shortest dist} = \frac{10.5}{\frac{1}{2} \times 3\sqrt{2}}$$

$$= \frac{7}{\sqrt{2}}$$

$$= \frac{7}{2}\sqrt{2}$$

7

(a)

$$\text{Grad } BC = \frac{8-6}{2-8} = -\frac{1}{3}$$

$$\text{Grad } AB = 3$$

(b)

$$\begin{aligned} \text{Let } A(0, y) \quad \frac{y-8}{0-2} &= 3 \\ y &= 2 \\ A \text{ is } (0, 2) \end{aligned}$$

(c)

$$\text{Grad of perpendicular bisector} = 3$$

$$\text{Midpoint } BC = \left(\frac{2+8}{2}, \frac{8+6}{2} \right) = (5, 7)$$

$$\text{Eqn of perpendicular bisector: } y - 7 = 3(x - 5)$$

$$y = 3x - 8$$

(d)

$$3y = 4x - 14$$

$$3(3x - 8) = 4x - 14$$

$$9x - 24 = 4x - 14$$

$$5x = 10$$

$$x = 2$$

$$y = 3(2) - 8 = -2$$

$$D(2, -2)$$

8

(a)

$$x^2 + y^2 - 12x - 8y - 13 = 0$$

$$x^2 - 12x + y^2 - 8y - 13 = 0$$

$$(x-6)^2 - 36 + (y-4)^2 - 16 - 13 = 0$$

$$(x-6)^2 + (y-4)^2 = 65$$

$$\text{Centre} = (6, 4) \quad \text{Radius} = \sqrt{65}$$

(b)

$$\text{Gradient of line} = -\frac{4}{7}$$

$$\text{Gradient of perpendicular} = \frac{7}{4}$$

$$\text{Equation of line: } y - 4 = \frac{7}{4}(x - 6)$$

$$y = \frac{7}{4}x - \frac{42}{4} + 4$$

$$y = \frac{7}{4}x - \frac{13}{2}$$

(c)

When the 2 lines meet, $y = \frac{7}{4}x - \frac{13}{2}$ ----(1)

and $4x + 7y = 117$ ---(2)

$$4x + 7\left(\frac{7}{4}x - \frac{13}{2}\right) = 117$$

$$4x + \frac{49}{4}x - \frac{91}{2} = 117$$

$$\frac{65}{4}x = \frac{325}{2}$$

$$x = 10, \quad 4(10) + 7y = 117$$

$$7y = 77$$

$$y = 11$$

Sub $(10, 11)$,

$$(x-6)^2 + (y-4)^2$$

$$= (10-6)^2 + (11-4)^2$$

$$= 16 + 49$$

$$= 65$$

The point $(10, 11)$ lie on the circle.

$4x + 7y = 117$ is a tangent to the circle and it touches the circle at $(10, 11)$.